Freshly

A pantry-first app that helps people scan groceries, track expiry dates, and prevent food waste by cooking, selling, or donating items before they expire.

Tagline: Scan it. Save it. Share it.

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Course: 3375

Section: #1

Group: Solo

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Requirements Gathering

Freshly is a pantry-first app where users scan groceries, track expiry dates, get reminders, and act (cook, sell, or donate) before food expires.

The purpose of this user study is to understand user habits and pain points around groceries and food waste and use that to shape requirements.

Objectives:

- Understand how often people grocery shop and how often food goes bad at home.
- Identify which food categories are wasted most often.
- Understand what people currently do to keep track of groceries and expiry dates (if anything).
- Evaluate whether scan-first input (with a quick manual correction option) is acceptable and realistically usable.
- Identify reminder preferences (timing and frequency), since reminders should help users stay aware without being excessive.
- Understand how likely users are to use the "cook / sell / donate" actions when items are close to expiring.

Choice of user study and user sample

- **User study type:** Online survey using Google Forms. The course guidance recommends mostly closed questions, so results are easier to analyze quantitatively.
- **Sample size:** 10-15
- **Target users:**
    - Students and working adults who shop weekly and sometimes forget items in the fridge.
    - Shared households/families where multiple people buy and use food.
    - Budget-conscious and eco-conscious users who want a realistic system that fits daily life.

Draft of user study survey

Section A: Background

1. **Which best describes you?**
    a. Student
    b. Working full time
    c. Working part time
    d. Not currently working

    e. Prefer not to say

2. **What is your household situation?**
    a. Live alone
    b. Roommates
    c. Partner
    d. Family

Section B: Grocery habits and waste

3. **How often do you go grocery shopping?**
    a. Multiple times per week
    b. Once per week
    c. Every 2 weeks
    d. Less often
4. **How often does food expire before you use it?**
    a. Multiple times per week
    b. About once per week
    c. A few times per month
    d. Rarely
    e. Almost never
5. **Which food types do you waste the most? (Select up to 3)**
    a. Fruits/vegetables
    b. Dairy
    c. Meat/seafood
    d. Leftovers
    e. Bread/baked goods
    f. Packaged snacks

Section C: Current tracking

6. **Do you currently track what groceries you have at home?**
    a. No
    b. In my head
    c. Notes app
    d. Spreadsheet
    e. Grocery/pantry app
7. **If you do not track groceries, what is the main reason?**
    a. Too much effort or time
    b. I forget to update it

c. Not useful for me
d. Shared household makes it harder

Section D: Scanning

8. **If you could add groceries by scanning instead of typing, how likely are you to use it? (1 = not likely, 5 = very likely)**
9. **If scanning does not correctly recognize an item sometimes, what would you prefer?**
    a. I am okay fixing it manually
    b. I am okay fixing it manually only if it is rare
    c. I would rather type items than fix scan mistakes
    d. I will not use scanning if it makes mistakes
10. **When would you want a reminder before an item expires?**
    a. Same day
    b. 1 day before
    c. 2 days before
    d. 3 days before
    e. 5 days before
    f. I do not want reminders
11. **What reminder frequency would you prefer?**
    a. Only when something is expiring soon
    b. Daily summary
    c. Weekly summary
    d. Never
12. **When food is close to expiring, what do you usually do?**
    a. Use it (cook/eat it)
    b. Give it to someone I know
    c. Throw it out
    d. It depends / no pattern
13. **How likely are you to use each option in an app when an item is close to expiring? (Scales of 1 – 5)**
    a. Cooking suggestions based on pantry items
    b. Selling locally
    c. Donating locally

Section E: Open ended

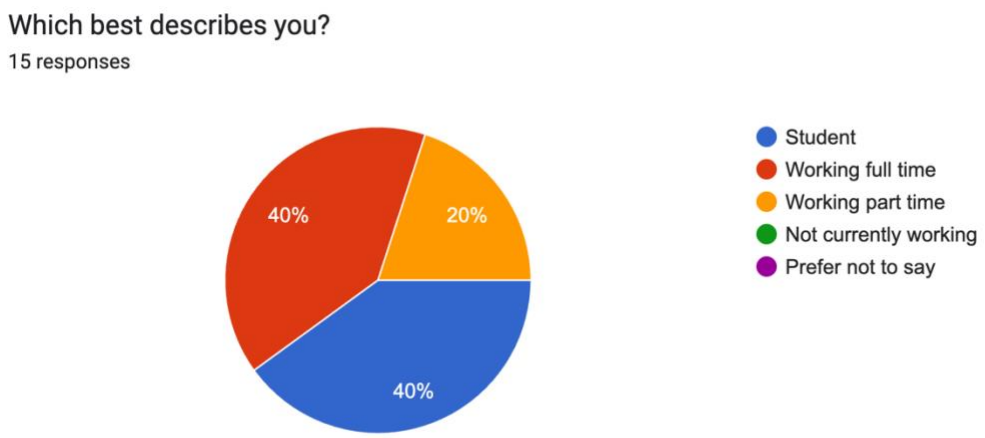
14. **What would make you trust local selling or donating more?**
15. **Is there anything else that would help you waste less food at home?**

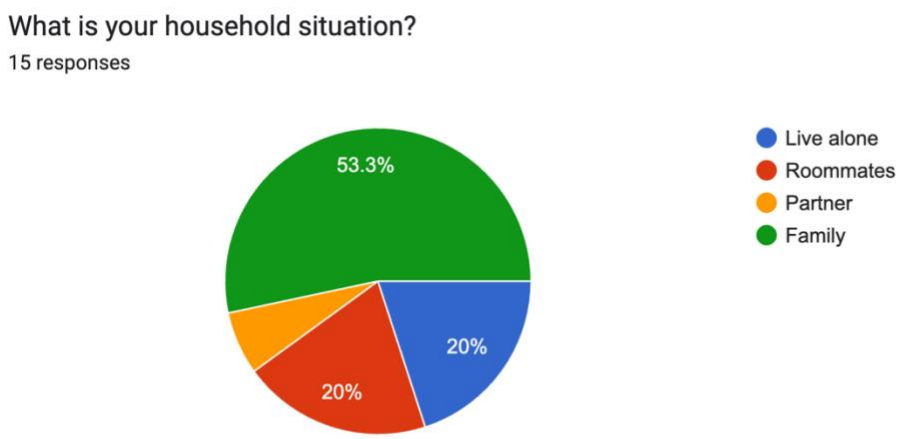
Data visualization and analysis:

The Freshly survey received 15 responses. Since most questions were closed ended, the results were analyzed by looking at response counts and comparing patterns across questions. For the rating-scale questions, the mean and median were also calculated. The open-ended responses were reviewed separately to see if any common ideas appeared.

Participant background

The respondents included 6 students, 6 full-time workers, and 3 part-time workers. For household situation, 8 lived with family, 3 lived alone, 3 lived with roommates, and 1 lived with a partner. This provided a mix of living situations that may affect grocery habits.



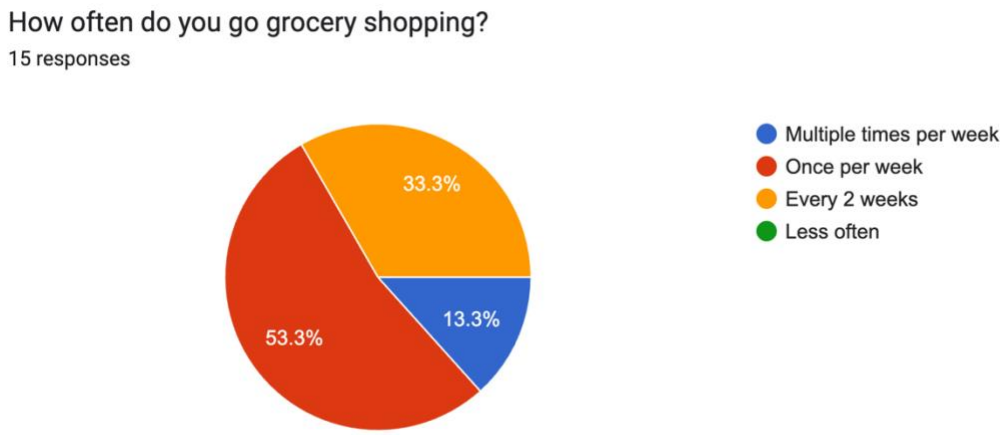


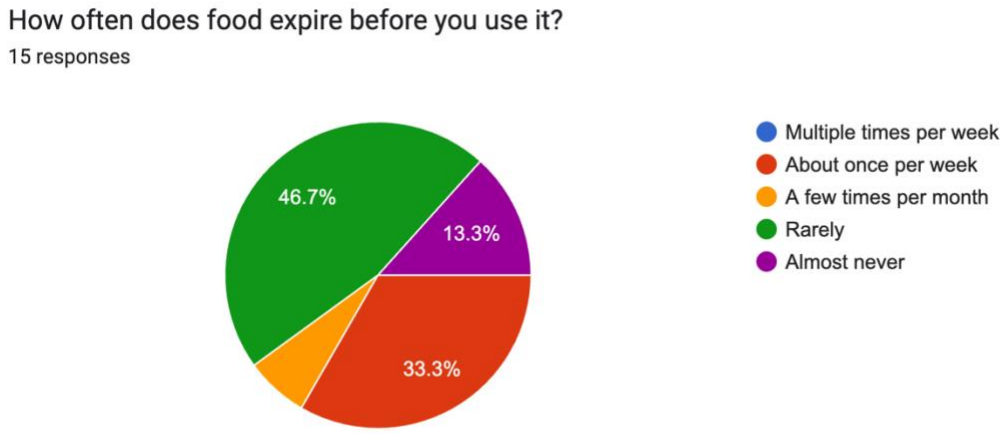
Grocery habits and food waste

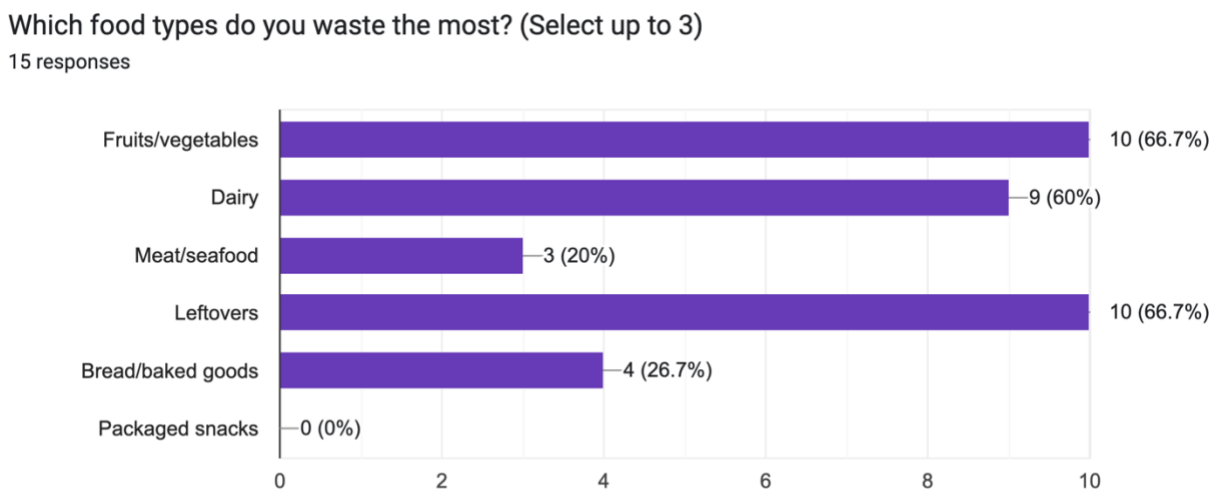
Most respondents shop for groceries once per week (8 responses). Another 5 respondents shop every two weeks, and 2 shop multiple times per week.

For food expiring before use, 7 respondents said it happens rarely, while 5 said about once per week, 1 said a few times per month, and 2 said almost never. This shows that food waste is still a recurring issue for some users.

The most wasted foods were fruits and vegetables (10 selections), leftovers (10), and dairy (9). Bread/baked goods appeared 4 times, meat/seafood 3 times, and packaged snacks were not selected.



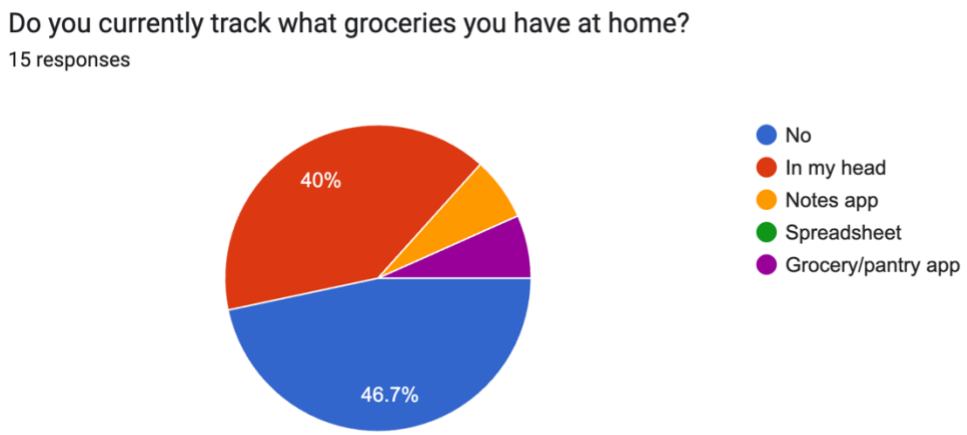


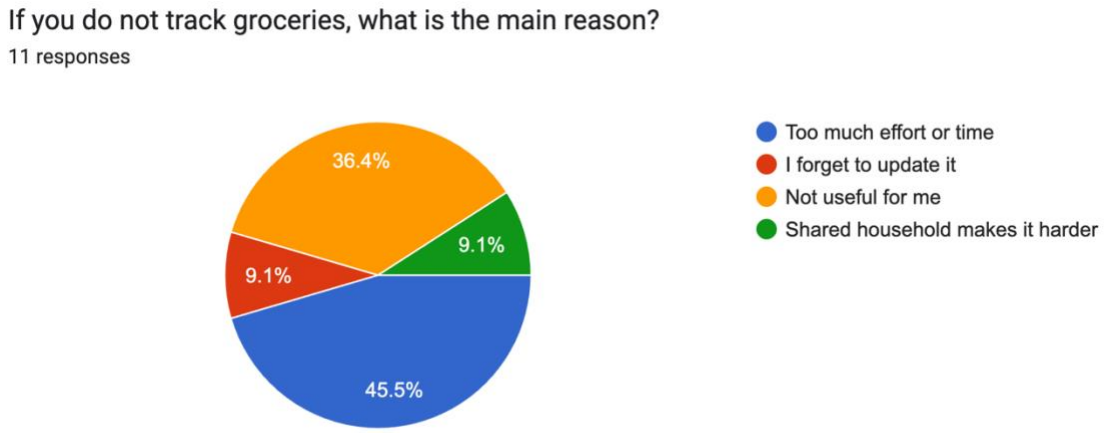


Current grocery tracking behavior

Most respondents do not use a structured tracking system. 7 said they do not track groceries, while 6 said they track in their head. Only one used a notes app and one used a grocery or pantry app.

The main reason for not tracking groceries was too much effort or time (5 responses), followed by not useful for me (4 responses). Forgetting to update and shared household difficulty were mentioned only once each.

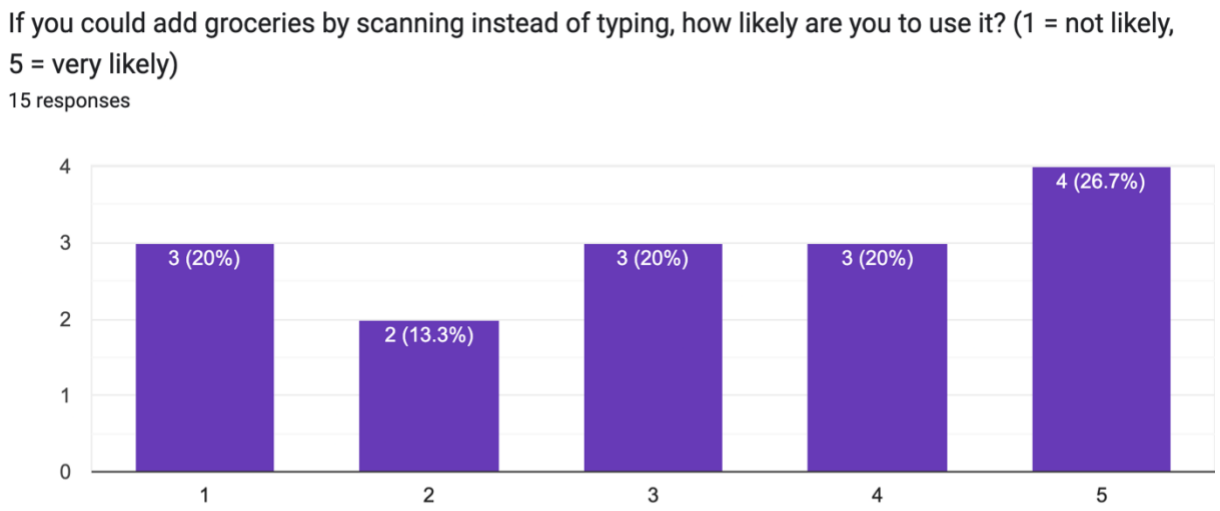


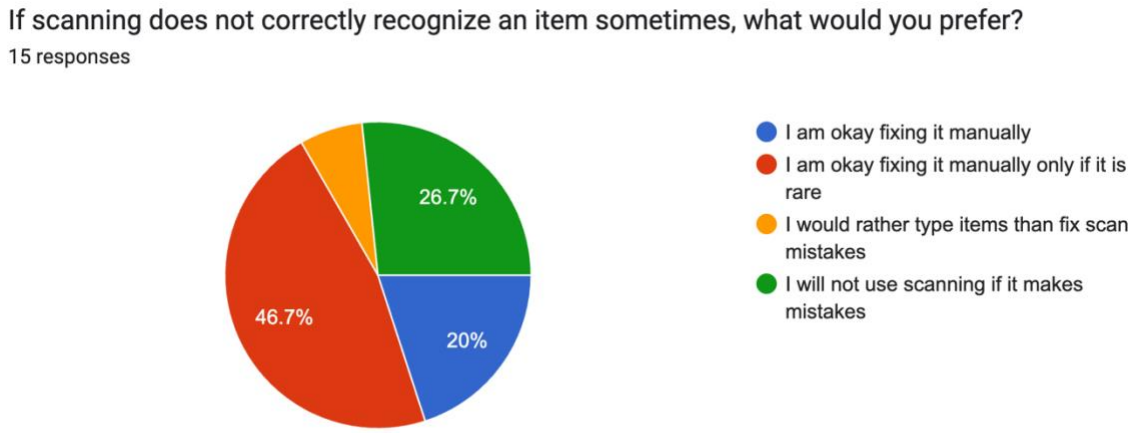


Scan input

Interest in scan-first input was mixed. The ratings were 1 = 3, 2 = 2, 3 = 3, 4 = 3, 5 = 4, giving a mean of 3.2 and a median of 3.

When asked about scan mistakes, 7 respondents said they would only tolerate errors if they are rare, 3 said they are okay fixing mistakes manually, 1 preferred typing items, and 4 said they would stop using scanning if it makes mistakes.





Reminder preferences

Reminder timing showed a clear preference for earlier notifications. Six respondents preferred reminders 5 days before expiry, and six preferred 3 days before expiry. Only two selected same day, and one selected 1 day before.

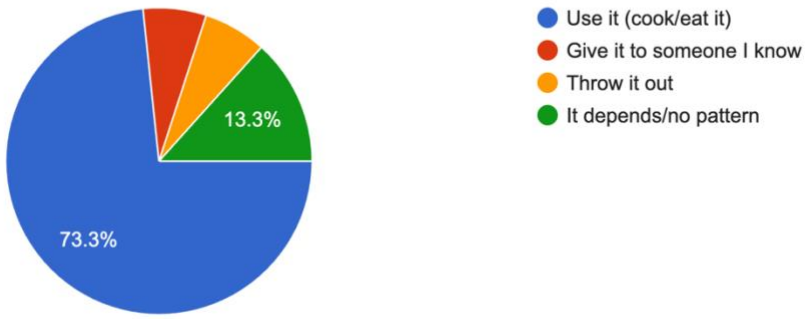
Actions when food is close to expiring

When food is close to expiring, 11 respondents said they use it, 2 said it depends, 1 gives it to someone, and 1 throws it out.

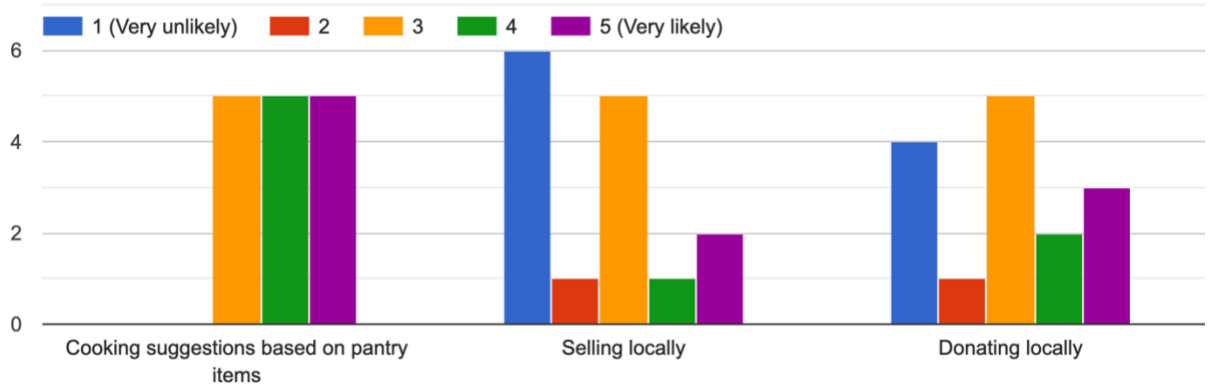
For feature ratings, cooking suggestions based on pantry items received the highest ratings, with a mean of 4.0. Selling locally received a mean of about 2.47, while donating locally received a mean of about 2.93.

When food is close to expiring, what do you do?

15 responses



How likely are you to use each option in an app when an item is close to expiring? (Scales of 1 – 5)



Open ended responses

Open-ended responses mostly mentioned convenience, clear item information, and trust as factors that would increase confidence in selling or donating food. Other suggestions included reminders, meal planning, and better visibility of items at home.

Insights from the Results

Several patterns appeared in the survey results:

- Food waste occurs occasionally for most respondents, though not always frequently.
- The foods most likely to be wasted are fruits/vegetables, leftovers, and dairy.
- Most users do not track groceries with a dedicated system.
- Effort is the main reason people do not track groceries.
- Scan-first input is acceptable but must be reliable.
- Users prefer reminders several days before expiry, especially 3–5 days.
- Cooking suggestions received the strongest support among the proposed features.
- Donation has moderate interest, while selling locally appears less appealing.

Impact on Freshly's requirements

The results suggest that Freshly should focus on helping users notice expiring food and use it in time. Features such as expiry reminders and cooking suggestions should be prioritized. Scan-first input can still be included, but it should not be the only way to add items. Donation could remain a secondary feature, while selling locally appears less important based on the survey results.

Requirements generation

Personas and Scenarios

1. Students with roommates

   Name: Neah

   Age: 21

   Situation: University student living with 2 roommates

   Aisha shares a fridge with her roommates. Groceries often get mixed together, and she sometimes forgets what food she already has. Because of

this, vegetables and leftovers occasionally expire before she remembers to use them.

Scenario:

After grocery shopping, Aisha adds a few items to the app using the scan feature. A few days later, the app sends a reminder that some vegetables will expire soon. She checks the app and sees cooking suggestions based on the items in her pantry, which helps her decide what to cook that evening.

2. Environmentally conscious user

   Name: Nina

   Age: 28
   Situation: Works full time and tries to reduce food waste.

   Nina is interested in making more sustainable choices and does not like throwing away food. However, she sometimes forgets about leftovers or produce in the fridge.

   Scenario

   Nina checks the app to see what items will expire soon. The app suggests recipes based on ingredients already in her pantry. She uses one of the suggestions to cook a meal using ingredients that might otherwise have gone to waste.

3. Busy working professional

   Name: Peter
   Age: 34
   Situation: Full-time worker who lives alone.

   Mark usually shops for groceries once a week. Because of his work schedule, he sometimes forgets about items in the fridge until they are close to expiring.

   Scenario

   One evening, Mark receives a notification that some dairy products will expire in three days. He opens the app and checks which items are affected. The reminder helps him plan his meals so the food gets used before it goes bad.

4. Budget-conscious household member

   Name: Atharva
   Age: 40
   Situation: Lives with family and often buys groceries in larger quantities.

   Because groceries are bought in bulk, some items occasionally expire before they are used. Sam wants to reduce waste and save money.

   Scenario

   The app shows that some packaged food will expire soon. Instead of throwing it away, Sam decides to donate it through the app. The app provides simple information about donation options and how to share the item locally.

Requirements generation

- The system should allow users to **add groceries quickly**, since users like Aisha and Mark do not want to spend much time tracking items.
- The app should maintain a **digital pantry view** so users can easily see what food is currently available at home.
- The system should track **expiry dates** to help users notice items that may be forgotten in the fridge.
- The app should provide **reminders before items expire**, which helps users like Mark who often forget about groceries during busy weeks.
- The system should highlight **items that are close to expiring**, making them easier to notice and prioritize.
- The app should provide **cooking suggestions using existing ingredients**, supporting users like Nina who want to reduce waste.
- The system should allow users to **donate or share food** that cannot be used in time, which supports situations like Sam's.
- The interface should remain **simple and low effort**, since the survey and personas both suggest that users avoid systems that feel complicated.

Synthesis of Requirements Gathering and Generation

This section combines the survey findings with the personas and scenarios to identify the main requirements for Freshly.

The survey showed that many users do not track groceries in a structured way, and the personas showed similar problems with forgetting what food is already at home. Because of this, Freshly should provide a **simple digital pantry** and an easy way to add items.

Another important result was that users avoid tracking groceries when it feels like too much effort. This means the app should support **quick item entry**, including scan input with manual entry as a fallback.

The foods most often wasted were **fruits/vegetables, leftovers, and dairy**, so the app should make **soon-to-expire items easy to notice**.

Users also preferred reminders **3 to 5 days before expiry**, which supports adding **early expiry reminders**.

The strongest feature result was **cooking suggestions based on pantry items**, so this should be treated as a **core feature**.

Finally, **donation had more support than selling**, so donation can be included as a secondary feature, while selling should be lower priority.

Final Requirements

High Priority

- Quick grocery entry
- Digital pantry view
- Expiry date tracking
- Highlighting soon-to-expire items
- Reminders several days before expiry
- Cooking suggestions based on pantry items

Medium Priority

- Donation or sharing options
- Simple pantry management

Lower Priority

- Selling food locally

Overall, the synthesis shows that Freshly should mainly focus on **helping users notice food early and use it before it gets wasted**.

AI Use Section

Table of AI Tools and Specific Use

AI Tool Name	Version / Account Type	Specific feature for which the AI tool was used
ChatGPT	GPT-5.4, ChatGPT Plus	Used to review and simplify survey question wording so the answer choices were clearer and easier to quantify.
ChatGPT	GPT-5.4, ChatGPT Plus	Used to suggest a structure for presenting survey results, including where to place graphs, counts, and short analysis.
ChatGPT	GPT-5.4, ChatGPT Plus	Used to shorten and revise some report sections so the writing sounded clearer and more natural.

Value Addition

The main ideas, survey design decisions, personas, interpretation of results, and final requirements were done by me and my group based on our own project and collected survey data. AI was only used as a support tool for wording, structure, and editing. I reviewed, selected, and revised the output before adding it to the report, and I adjusted the writing to fit the assignment requirements and our project context.

Appendix

- “does this survey question sound clear for a google form survey? trying to avoid answers like ‘other’ since i need results i can quantify”
- “i have 15 survey responses. how should i structure the results and analysis section for a report?”
- “do these requirements look reasonable based on the survey results?”

Work log

Date	Number of Hours	Description of Work Done
Jan 24, 2026	2.0	Defined initial concept as a local marketplace for expiring household food. Wrote problem statement and target user notes.
Jan 25, 2026	2.0	Broke down marketplace-first features and drafted original experience loop concept.
Jan 26, 2026	1.5	Conducted competitor research on Too Good To Go, OLIO, and pantry tracking apps.
Jan 27, 2026	1.0	Reframed app around minimizing food waste after instructor feedback. Updated core flow.
Feb 3, 2026	1.5	Researched and practiced Figma via tutorials.
Feb 6, 2026	2.0	Created mobile prototype frames and layout structure in Figma.
Feb 13, 2026	2.0	Recorded and created the Figma tutorial video.
Feb 17, 2026	1.0	Edited the Figma tutorial video for clarity.
March 5, 2026	2.0	Drafted survey questions and created the Google Form for requirements gathering.
March 6, 2026	0.5	Shared survey and started collecting responses.
March 7, 2026	0.5	Continued monitoring survey responses.
March 8, 2026	2.5	Exported survey results, analyzed data, and wrote the results and insights section.
March 9, 2026	3	Created personas and scenarios based on survey results. Generated requirement lists.

		Completed AI use section, appendix prompts, and final formatting of the document.

Reference:

- https://www.nngroup.com/articles/scenario-mapping-personas